날씨 앱 프로젝트 (9개 코드 파일)

앱 기능

- 일주일 날씨 표시
- 전국 지역별 날씨 조회
- 일출 / 일몰 시간 표시
- 날씨 기반 옷차림 추천
- 날씨 이미지 표시
- 음성(TTS)으로 날씨 안내
- 반응형 UI

사용 기술:

- React + Vite
- Tailwind CSS
- OpenWeather API
- Web Speech API

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1. package.json

```
{
  "name": "weather-fashion-app",
  "private": true,
  "version": "1.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build"
  },
  "dependencies": {
    "react": "^18.2.0",
    "react-dom": "^18.2.0",
    "lucide-react": "^0.379.0"
  },
  "devDependencies": {
    "vite": "^5.2.0",
    "tailwindcss": "^3.4.1",
    "autoprefixer": "^10.4.19",
    "postcss": "^8.4.38"
  }
}
```

---

2. src/main.jsx

```
import React from 'react'
import ReactDOM from 'react-dom/client'
import App from './App'
import './index.css'

ReactDOM.createRoot(document.getElementById('root')).render(
 <React.StrictMode>
  <App />
 </React.StrictMode>
)
```

---

3. src/App.jsx

```
import { useEffect, useState } from 'react'
import Header from './components/Header'
import WeatherCard from './components/WeatherCard'
import WeeklyForecast from './components/WeeklyForecast'
import ClothingTip from './components/ClothingTip'
import VoiceWeather from './components/VoiceWeather'

const API_KEY = 'YOUR_API_KEY'

export default function App() {
 const [weather, setWeather] = useState(null)
 const [forecast, setForecast] = useState([])
 const [city, setCity] = useState('Busan')

 useEffect(() => {
  fetchWeather()
 }, [city])

 const fetchWeather = async () => {
  const weatherRes = await fetch(
   `https://api.openweathermap.org/data/2.5/weather?q=${city}&appid=${API_KEY}&units=metric`
  )

  const weatherData = await weatherRes.json()
  setWeather(weatherData)

  const forecastRes = await fetch(
   `https://api.openweathermap.org/data/2.5/forecast?q=${city}&appid=${API_KEY}&units=metric`
```

```
    )

    const forecastData = await forecastRes.json()
    setForecast(forecastData.list.slice(0, 7))
  }

  return (
    <div className="min-h-screen bg-sky-100 p-5">
      <Header city={city} setCity={setCity} />

      {weather && (
        <>
          <WeatherCard weather={weather} />
          <ClothingTip temp={weather.main.temp} />
          <VoiceWeather weather={weather} />
        </>
      )}

      <WeeklyForecast forecast={forecast} />
    </div>
  )
}
```

---

4. src/components/Header.jsx

```
export default function Header({ city, setCity }) {
  return (
    <div className="bg-white rounded-3xl shadow-xl p-5 mb-5 flex gap-3 items-center">
      <h1 className="text-3xl font-bold text-sky-600">
        스마트 날씨 앱
      </h1>

      <input
        value={city}
        onChange={(e) => setCity(e.target.value)}
        className="border rounded-xl px-4 py-2"
        placeholder="도시 입력"
      />
    </div>
  )
}
```

---

5. src/components/WeatherCard.jsx

```
export default function WeatherCard({ weather }) {
  const sunrise = new Date(weather.sys.sunrise * 1000)
  const sunset = new Date(weather.sys.sunset * 1000)

  return (
    <div className="bg-white rounded-3xl shadow-xl p-6 mb-5">
      <div className="flex items-center justify-between">
        <div>
          <h2 className="text-4xl font-bold">
            {weather.name}
          </h2>

          <p className="text-5xl mt-3 font-bold text-sky-600">
            {Math.round(weather.main.temp)}°C
          </p>

          <p className="text-gray-600 text-xl mt-2 capitalize">
            {weather.weather[0].description}
          </p>
        </div>

        <img
          src={`https://openweathermap.org/img/wn/${weather.weather[0].icon}@4x.png`}
          alt="weather"
          className="w-40"
        />
      </div>

      <div className="grid grid-cols-2 gap-4 mt-5 text-lg">
        <div className="bg-sky-50 p-4 rounded-2xl">
          🌅 일출: {sunrise.toLocaleTimeString()}
        </div>

        <div className="bg-orange-50 p-4 rounded-2xl">
          🌇 일몰: {sunset.toLocaleTimeString()}
        </div>
      </div>
    </div>
  )
}
```

6. src/components/WeeklyForecast.jsx

```
export default function WeeklyForecast({ forecast }) {
 return (
  <div className="bg-white rounded-3xl shadow-xl p-6 mt-5">
   <h2 className="text-2xl font-bold mb-5">
    일주일 날씨
   </h2>

   <div className="grid md:grid-cols-4 gap-4">
    {forecast.map((item, index) => (
     <div
      key={index}
      className="bg-sky-50 rounded-2xl p-4 text-center"
     >
      <p>
       {new Date(item.dt * 1000).toLocaleDateString()}
      </p>

      <img
       src={`https://openweathermap.org/img/wn/${item.weather[0].icon}@2x.png`}
       alt="weather"
       className="mx-auto"
      />

      <p className="text-2xl font-bold">
       {Math.round(item.main.temp)}°C
      </p>
     </div>
    ))}
   </div>
  </div>
 )
}
```

7. src/components/ClothingTip.jsx

```
export default function ClothingTip({ temp }) {
 let tip = ''

 if (temp >= 28) {
  tip = '반팔, 반바지, 시원한 옷 추천'
 } else if (temp >= 20) {
```

```
    tip = '얇은 긴팔과 가디건 추천'
  } else if (temp >= 10) {
    tip = '자켓과 긴바지 추천'
  } else {
    tip = '두꺼운 외투와 목도리 추천'
  }

  return (
    <div className="bg-white rounded-3xl shadow-xl p-6 mt-5">
      <h2 className="text-2xl font-bold mb-3">
        오늘의 옷차림
      </h2>

      <p className="text-xl text-gray-700">
        👕 {tip}
      </p>
    </div>
  )
}
```

---

8. src/components/VoiceWeather.jsx

```
export default function VoiceWeather({ weather }) {
  const speakWeather = () => {
    const text = `${weather.name}의 현재 기온은 ${Math.round(
      weather.main.temp
    )}도이고 날씨는 ${weather.weather[0].description} 입니다.`

    const utterance = new SpeechSynthesisUtterance(text)
    utterance.lang = 'ko-KR'

    speechSynthesis.speak(utterance)
  }

  return (
    <button
      onClick={speakWeather}
      className="bg-sky-500 hover:bg-sky-600 text-white px-6 py-4 rounded-2xl mt-5 text-lg font-
bold"
    >
      🔊 음성으로 날씨 듣기
    </button>
  )
}
```

9. src/index.css

```
@tailwind base;
@tailwind components;
@tailwind utilities;

body {
  margin: 0;
  font-family: sans-serif;
  background: #dbeafe;
}
```

추가 디자인 아이디어

메인 화면 디자인

- 하늘색 그라데이션 배경
- 날씨 상태에 따라 배경 변경
- 맑음 → 밝은 하늘
- 비 → 어두운 회색
- 눈 → 흰색 톤

애니메이션

- 구름 이동 효과
- 비 내리는 효과
- 해 뜨는 애니메이션

추가 가능 기능

- 현재 위치 자동 감지
- 전국 도시 즐겨찾기
- 습도 / 바람 표시
- 미세먼지 표시
- 다크모드
- AI 코디 추천
- 음성 명령 검색

실행 방법

```
npm install
npm run dev
```

OpenWeather API 발급

1. OpenWeather 사이트 가입
2. API KEY 발급
3. App.jsx의 YOUR_API_KEY 변경

```
const API_KEY = '발급받은키'
```